

PHYS 357: Honours Quantum Physics 1

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Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital Angular Momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 2 (Angular Momentum of a Rigid Body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega, \quad (2)$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment

Definition 3 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

¹A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z, \quad (3)$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = \frac{q}{T} = \frac{qv}{2\pi r}$ because it passes any given point on the loop once per period.

Result 1 (Gyromagnetic Ratio). *The magnetic moment is proportional to the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \quad \gamma = \frac{q}{2mc}. \quad (4)$$

The constant γ is the gyromagnetic ratio.

Remark 2. *This is a purely classical result, in the sense that it follows from Newtonian mechanics and classical electromagnetism alone. The charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

Definition 4 (Stern-Gerlach Apparatus). *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient exerts a force $F_z = \mu_z \partial B_z / \partial z$, so each atom is deflected by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma \vec{S}$, the position on the screen is a direct measurement of S_z .*

2.2 Spin and the magnetic moment

In quantum mechanics the intrinsic angular momentum $\vec{S}$, called spin, replaces the orbital angular momentum $\vec{L}$. The magnetic moment is still proportional to it, but with an extra dimensionless factor,

$$\vec{\mu} = \frac{qg}{2mc} \vec{S}, \quad (5)$$

where g is the g-factor of the particle. It has no classical analogue and must be measured or derived from relativistic quantum theory.

Definition 5 (Gyromagnetic ratio). *The gyromagnetic ratio is the ratio of the magnetic moment to the angular momentum,*

$$\gamma = \frac{qg}{2mc}, \quad (6)$$

where q is the charge, g is the g-factor, m is the mass, and c is the speed of light.

²Here c is in Gaussian units.

2.2.1 Magnetons

Definition 6 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (7)$$

It carries units of angular momentum, so $[\hbar] = [L_z]$.

Factoring $\hbar$ out of the gyromagnetic ratio gives a natural unit of magnetic moment for each mass scale.

Definition 7 (Bohr magneton).

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (8)$$

Definition 8 (Nuclear magneton).

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (9)$$

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (10)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

2.2.2 Values for the three particles

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. *The neutron is neutral but still has a magnetic moment, so $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 4. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

2.3 The apparatus: force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (11)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U, \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}. \quad (12)$$

Remark 5. *A uniform field gives no force, only a torque. The Stern-Gerlach apparatus needs a field gradient to deflect anything.*

2.3.1 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z. \quad (13)$$

Taking the field along z ,

$$U = -\mu_z B_z = g_e \frac{|e|}{2m_e c} S_z B_z, \quad (14)$$

and therefore

$$F_z = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z, \quad (15)$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 2.

$$F_z \propto S_z. \quad (16)$$

The deflection of an atom measures the z component of its spin directly.

2.4 The result: quantization of S_z

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta. \quad (17)$$

Classically $\theta \in [0, \pi]$, so $\cos \theta \in [-1, 1]$ and S_z varies continuously over $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, the beam should smear into a continuous band. It does not.

Result 3 (Quantization of spin). *The electron spin angular momentum is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (18)$$

The beam splits into two spots.

2.4.1 Ket notation

New notation, a “ket”:

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The spin of an electron is a two-level system, so it can be represented as a two-dimensional vector space. The spin state of an electron can be represented as a linear combination of the basis states $|\uparrow\rangle$ and $|\downarrow\rangle$:

A Appendix

B Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>